

AUDIT DELIVERABLE

STATEFLUX REAL IPSEC LAB VALIDATION REPORT

Empirical, cryptographically grounded evaluation of IPsec configurations, negotiation space frontiers, and deterministic change outcomes.

Generated Date:	29 September 2026
Scope:	40 IPsec Tunnels • 20 Gateway Endpoints
Validation Environment:	strongSwan 5.9.8 • Libreswan 4.12
Report Identifier:	rep-lab-20260929145158

1. Executive Summary

PURPOSE	Validate whether STATEFLUX's IPsec change-impact predictions agree with observed behavior in a controlled real-IPsec testbed.
VALIDATION	5 controlled scenarios
RESULT	5/5 controlled testbed scenarios matched
ENVIRONMENT	strongSwan 5.9.8, Libreswan 4.12
LIMITATION	Controlled testbed validation measures prediction agreement in isolated containers; does not claim broad production ML accuracy.

2. Validation Metric Summary

5	5	0	strongSwan	Libreswan
SCENARIOS	MATCHED	MISMATCHES	v5.9.8	v4.12

3. Empirical Scenario Validation Table

Scenario	Description	Predicted	Observed	Match	Key Observation
LAB-01	Secure Suite-B Baseline (strongSwan <-> strongSwan)	ESTABLISHED	ESTABLISHED	YES	AES-GCM / DH20 / PFS
LAB-02	Weak but Functional Fallback (strongSwan <-> strongSwan)	ESTABLISHED	ESTABLISHED	YES	Weak suite accepted
LAB-03	Encryption Algorithm Incompatibility (strongSwan <-> Libreswan)	FAILED	FAILED	YES	NO_PROPOSAL_CHOSEN
LAB-04	Diffie-Hellman Group Incompatibility (strongSwan <-> strongSwan)	FAILED	FAILED	YES	DH mismatch
LAB-05	Latent Rekey Failure (strongSwan <-> Libreswan)	LATENT_FAILURE	LATENT_FAILURE	YES	NO_PROPOSAL_CHOSEN during rekey

4. LAB-05 — Latent Rekey Failure Deep Dive

LAB-05 demonstrates the most hazardous condition in enterprise IPsec networks: policy updates appear completely successful upon execution, but cause total connectivity loss during subsequent RFC 7296 cryptographic rekeying.

Step 1	INITIAL STATE: CHILD_SA ESTABLISHED	ESTABLISHED
Step 2	POLICY UPDATE APPLIED: RESPONDER REMOVES FALLBACK CIPHER	DORMANT BUG
Step 3	WAIT FOR REKEY → CREATE_CHILD_SA INITIATED	REKEY TRIGGER
Step 4	NO_PROPOSAL_CHOSEN RETURNED BY RESPONDER	REKEY REJECTED
Step 5	REKEY FAILURE → LATENT_FAILURE CONFIRMED	DROPPED

PREDICTION	ACTUAL OBSERVATION	VALIDATION
LATENT_FAILURE	CHILD_SA initially established. Rekey attempt failed with NO_PROPOSAL_CHOSEN.	PREDICTION = OBSERVATION

5. Cryptographic Evidence Chain

Traceable provenance linking configurations to empirical lab capture frames:

Evidence ID	Stage	Source Artifact	Fact Description
E-001	Configuration	data/seed/endpoints.json	Peer configured with legacy fallback proposal AES-128-CBC
E-005	Negotiation	app.services.twin_service	Dynamic Security Floor identifies dormant downgrade vulnerability
E-008	Simulation	app.services.simulation_engine	What-if engine predicts LATENT_FAILURE at 30s rekey interval
E-013	Lab Log	lab/logs/lab05_charon.log	CREATE_CHILD_SA rejected with NO_PROPOSAL_CHOSEN at 00:30.030
E-014	PCAP	lab/captures/lab05_latent_rekey_failure.pcap	Frame #14 contains IKEv2 Notify payload code 14

6. Technical Methodology & Environment

Controlled testbed topologies isolate strongSwan 5.9.8 and Libreswan 4.12 daemons in dedicated Linux network namespaces, using veth interconnects to capture raw traffic on port 500/4500 and parse charon/pluto syslog streams at debug loglevels.